JACOB PFAU

UK AISI Alignment Research Lead · jacob.pfau@gmail.com

EDUCATION

New York University, He and Bowman Labs Fall 2022 - Present
PhD Student, Center for Data Science

University of Edinburgh Sept. 2021 - Aug. 2022
MSc Mind, Language and Embodied Cognition (Philosophy Department). First class honors.

Ecole Polytechnique (Université Paris-Saclay), Paris Sept. 2018 - Aug. 2019
M1, Masters Computer Science, Data Science track.

Amherst College Sept. 2013 - May 2017
BA Mathematics & 5 College Logic Certificate. ETH Zurich Visiting student.

EMPLOYMENT

UK AI Security Institute, Alignment Team

Research Lead 2026–Present
Co-lead team of 16 developing AGI-ready alignment methods: setting research vision, steering academic and industry lab collaborations, and collaborating hands-on across team projects spanning scalable oversight, automated alignment, personas, and learning theory.

Research Scientist 2025–2026
Research focused on scalable oversight: published debate safety case; ongoing work on adversarial RL methods and conceptual alignment work. Grew alignment team from 3 to 16 staff. Co-led £25M grants round engaging leading researchers in CS theory, economics, and ML to develop new alignment research agendas; co-organized alignment conference with similar goals.

MATS, *Mentor* Jan. 2026 - Present
Mentoring two researchers on empirical study of debate as a scalable oversight method, using RL to train LLMs to debate.

LASR: London AI Safety Research Labs, *Mentor* July 2025 - Jan. 2026
Mentored team of four researchers working on empirically studying the unexploitable search problem for language models. This involves (1) constructing model organisms: misaligned LMs with hidden objectives (2) iterating on methods for training away such hidden objectives.

Anthropic, *Contractor* Dec. 2023 - May 2024
Collaboration commissioned by Anthropic to assess the implications of theories of welfare and consciousness on LLMs; coauthored 'Taking AI Welfare Seriously'.

AI Safety Camp, *Mentor* Mar. 2023 - July 2023
Mentored four researchers on project developing a benchmark for evaluating LM self-consistency.

University of California, San Francisco. Keiser Lab, *Bold and Basic Fellow* Mar. 2019 - Aug. 2020
Developed interpretability methods and benchmarking datasets for clinical imaging.

PUBLICATIONS

An Alignment Safety Case Sketch Based on Debate. arXiv,
Buhl M D, Pfau J, Hilton B, Irving G. *arXiv Preprint* (2025)

When Benchmarks Talk: Re-Evaluating Code LLMs with Interactive Feedback. DOI,
Pan J, Shar R, Pfau J, Talwalkar A, He H, Chen V. *Findings of the Association for Computational Linguistics: ACL* (2025)

Let's Think Dot by Dot: Hidden Computation in Transformer Language Models. arXiv,
Pfau J, Merrill W, Bowman S. *COLM* (2024)

Taking AI Welfare Seriously. arXiv,
Long R, Sebo J, Butlin P, Finlinson K, Fish K, Harding J, Pfau J, Sims T, Birch J, Chalmers D *arXiv Preprint* (2024)

Steering Without Side Effects: Improving Post-Deployment Control of Language Models. arXiv,
Stickland A C, Lyzhov A, Pfau J, Mahdi S, Bowman S R *arXiv Preprint* (2024)

Eliciting Language Model Behaviors using Reverse Language Models. OpenReview,
Pfau J, Infanger A, Sheshadri A, Panda A, Huebner C, Michael J. *Socially Responsible Language Modelling Research (SoLaR) Workshop at NeurIPS* (2023)

Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback. arXiv,
Casper S, ... Pfau J, et al. *Transactions on Machine Learning Research (TMLR)* (2023)

Self-Consistency of Large Language Models under Ambiguity. DOI,
Bartsch H, ... Pfau J. *Proceedings of the 6th BlackboxNLP Workshop* (2023)

Goal misgeneralization in deep reinforcement learning. arXiv,
Langosco L, Koch J, Le J, Sharkey L, Pfau J, Krueger D. *ICML* (2022)

Robust Semantic Interpretability: Revisiting Concept Activation Vectors. arXiv,
Pfau J, Young A, Wei J, Wei M, Keiser M. *ICML Workshop on Human Interpretability* (2020)

Stress Testing Reveals Gaps in Clinic Readiness of Image-Based Diagnostic AI Models. DOI,
Young A, Fernandez K, Pfau J, et al. *npj Digital Medicine* (2021)

Artificial Intelligence in Dermatology: A Primer. DOI,
Young A, ... Pfau J, et al. *Journal of Investigative Dermatology* (2020)

Artificial Intelligence in Teledermatology. DOI,
Xiong M, Pfau J, et al. *Current Dermatology Reports* (2019)

Global Saliency: Aggregating Saliency Maps to Assess Dataset Artefact Bias. arXiv,
Pfau J, Young A, Wei M, Keiser M. *Machine Learning for Health (ML4H) at NeurIPS* (2019)

FELLOWSHIPS AND AWARDS

OpenAI Superalignment PhD Fellowship, *New York University* Sept. 2024 - June. 2025
Awarded PhD fellowship studying LLM consistency methods to scalable oversight problems.

Bold and Basic Fellowship, *University of California, San Francisco* Sept. 2019 - Aug. 2020
Awarded research fellowship for applying CNNs to skin cancer image and genomic data.

Research Internship Prize, *Ecole Polytechnique, Paris* Oct. 2019

Labex DigiCosme Fellowship, *Labex DigiCosme, Paris* Sept. 2018 - Mar. 2019

SKILLS AND MISCELLANEOUS

LANGUAGES: English (native), German (fluent), French (conversant), Mandarin (basic)